Supplementary Materials:
Circuit Motif Spectroscopy

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S1 Extended Methods

S1.1 Null Model Construction

The null models are generated using the Goni Method 1 edge-swap algorithm, which preserves in-degree and out-degree sequences while randomizing connectivity. The algorithm proceeds as follows:

1. Initialize with the original directed acyclic graph (DAG).
2. For each swap iteration (total swaps $= 10 \times |E|$):
   (a) Select two random edges (u, v) and (x, y).
   (b) Propose swap: replace with (u, y) and (x, v).
   (c) Accept swap only if the resulting graph remains acyclic (verified via topological sort).
3. Output the rewired graph as a null model.

This procedure ensures that each null model preserves the degree distribution while randomizing higher-order structure, making it appropriate for testing motif enrichment.

S1.2 Pruning Threshold Sensitivity

Table 1 shows Spearman ρ correlation matrices between Z-scores computed at different pruning thresholds (60th, 75th, 90th percentile).

Table 1: Pruning threshold stability: Spearman ρ between Z-scores at different thresholds.

Motif		60th	75th	90th
021U	60th	1.000	0.499	0.288
	75th	0.499	1.000	0.401
	90th	0.288	0.401	1.000
021C	60th	1.000	0.499	0.288
	75th	0.499	1.000	0.401
	90th	0.288	0.401	1.000
021D	60th	1.000	0.499	0.288
	75th	0.499	1.000	0.401
	90th	0.288	0.401	1.000
030T	60th	1.000	0.499	0.288
	75th	0.499	1.000	0.401
	90th	0.288	0.401	1.000

S1.3 Convergence Diagnostics

Median convergence is reached at $N = 30$ null models. Table 2 shows the fraction of graphs converged at each checkpoint.

S1.4 Computational Cost

S2 Full Evidence Table

Table 4 presents the complete master evidence table with all 59 evidence rows spanning hypotheses H1–H5.

Table 2: Convergence diagnostics: fraction of graphs with Z-score relative error $< 5\%$.

Null Count	Fraction Converged
$N = 5$	0.10
$N = 10$	0.15
$N = 20$	0.45
$N = 30$	0.70
$N = 40$	0.85
$N = 50$	1.00

Table 3: Runtime per experiment phase.

Experiment	Runtime (s)	Runtime (min)
Corpus-Level Stats (exp1_it6)	1777.5	29.6
Node Ablation (exp2_it4)	0.0	0.0
Failure Prediction (exp3_it4)	1440.6	24.0
Z-Score Fortification (exp3_it5)	0.0	0.0
4-Node Characterization (exp2_it5)	34.1	0.6
Total	**3252.2**	**54.2**

Table 4: Master Evidence Table: all 59 evidence rows across hypotheses H1–H5.

#	Hyp	Sub-claim	Primary Metric	Value	95% CI	p-value	Effect Size	Criterion	Met
1	H1	FFL (030T) overrepresented across all attr...	FFL mean Z-score (200 graphs, 50 null...	46.2066	—	—	Z-score=46.207	Mean Z > 2	✓
2	H1	FFL overrepresentation in antonym domain	FFL median Z-score (antonym)	26.2773	—	—	Z-score=26.277	Domain median Z > 2	✓
3	H1	FFL overrepresentation in arithmetic domain	FFL median Z-score (arithmetic)	55.3803	—	—	Z-score=55.380	Domain median Z > 2	✓
4	H1	FFL overrepresentation in code_completion ...	FFL median Z-score (code_completion)	44.9965	—	—	Z-score=44.996	Domain median Z > 2	✓
5	H1	FFL overrepresentation in country_capital ...	FFL median Z-score (country_capital)	47.0661	—	—	Z-score=47.066	Domain median Z > 2	✓
6	H1	FFL overrepresentation in multi_hop_reason...	FFL median Z-score (multi_hop_reasoning)	36.3839	—	—	Z-score=36.384	Domain median Z > 2	✓
7	H1	FFL overrepresentation in rhyme domain	FFL median Z-score (rhyme)	45.5278	—	—	Z-score=45.528	Domain median Z > 2	✓
8	H1	FFL overrepresentation in sentiment domain	FFL median Z-score (sentiment)	59.3607	—	—	Z-score=59.361	Domain median Z > 2	✓
9	H1	FFL overrepresentation in translation domain	FFL median Z-score (translation)	50.8267	—	—	Z-score=50.827	Domain median Z > 2	✓
10	H1	FFL overrepresentation survives layer-pres...	Layer-preserving null mean Z-score	56.8751	—	$< 10^{-300}$	Z-score=56.875	Layer-preserving ...	✓
11	H1	Individual graph FFL significance survives...	BH-FDR surviving tests (combined 3+4 ...	0	—	—	ratio=0	Some tests surviv...	×
12	H1	Z-score estimates converge with sufficient...	Median convergence null count (N)	30.0000	—	—	ratio=30.000	Convergence withi...	✓
13	H1	Z-score rankings stable across pruning thr...	Spearman rho (60th vs 75th pct pruning)	0.49900	—	—	ratio=0.4990	rho > 0.5	×
14	H1	3-node motif spectrum captures multi-dimen...	Effective dimensions in 3-node spectrum	1.0000	—	—	ratio=1.000	Multiple independ...	×
15	H2	Weighted motif features cluster by domain ...	Weighted NMI (k=4)	0.70500	—	—	NMI=0.7050	NMI > 0.3	✓
16	H2	Binary motif counts cluster by domain (k=8)	Binary NMI (k=8)	0.10100	—	—	NMI=0.1010	NMI > 0.3	×
17	H2	Combined motif+graph features cluster by d...	Combined NMI (k=8)	0.84400	—	—	NMI=0.8440	NMI > 0.3	✓

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#	Hyp	Sub-claim	Primary Metric	Value	95% CI	p-value	Effect Size	Criterion	Met
18	H2	Graph-stats features alone cluster by doma...	Graph-stats NMI (k=4)	0.62000	—	—	NMI=0.6200	NMI > 0.3	✓
19	H2	Count-ratio motif features cluster by domain	Count-ratio NMI (k=8, iter 2)	0.85100	—	—	NMI=0.8510	NMI > 0.3	✓
20	H2	Motif features carry unique variance not e...	Unique motif R-squared (McFadden deco...	0.01837	[0.0213, 0.0448]	—	R-square=0.0184	CI excludes 0	✓
21	H2	Graph statistics also carry unique variance	Unique graph-stat R-squared	0.05023	—	—	R-square=0.0502	Positive value	✓
22	H2	Most domain-predictive variance is shared ...	Shared R-squared	0.92987	—	—	R-square=0.9299	Informational	✓
23	H2	Motif residuals still cluster by domain af...	Residualized motif NMI (best k)	0.26430	—	0.001	NMI=0.2643	NMI > 0 with p < ...	✓
24	H2	Residualized clustering is statistically s...	Permutation test p-value (1000 perms)	0.00100	—	0.001	NMI=0.2643	p < 0.05	✓
25	H2	Motif and graph-stat spaces share signific...	CCA significant dimensions	7.0000	—	—	ratio=0.7000	Some dims signifi...	✓
26	H2	Motif clustering persists after domain-lev...	Domain-normalized motif NMI (k=8)	0.65940	—	—	NMI=0.6594	NMI comparable to...	✓
27	H2	FFL intensity strongly predicts domain mem...	FFL intensity eta-squared (ANOVA)	0.91700	—	—	eta-squa=0.9170	eta-squared > 0.1...	✓
28	H2	FFL path dominance predicts domain membership	FFL path dominance eta-squared	0.81000	—	—	eta-squa=0.8100	eta-squared > 0.1...	✓
29	H2	Mutual information retained after residual...	MI retained fraction (motif—gstat)	0.23620	—	—	ratio=0.2362	Some MI retained ...	✓
30	H3	Best model predicts failures better than c...	Best model AUC (graph_stats_only, log...	0.58284	[0.4840, 0.6684]	—	ratio=0.0828	AUC > 0.65	×
31	H3	Motif features alone predict failures	Motif-only AUC (logistic)	0.49574	—	—	ratio=-0.0043	AUC > 0.5	×
32	H3	Adding motifs to baseline improves prediction	Full model AUC (logistic)	0.55407	—	—	ratio=0.0541	AUC > domain+grap...	×
33	H3	Baseline domain+graph model performance	Domain+graph AUC (logistic)	0.57448	—	—	ratio=0.0745	Baseline reference	✓
34	H3	Motif features significantly improve over ...	Motif lift bootstrap p-value	0.86500	[-0.0586, 0.0160]	0.865	ratio=0	p < 0.05	×
35	H3	Within-domain (code_completion) motif fail...	Motif-only AUC (code_completion)	0.52101	—	—	ratio=0.0210	AUC > 0.5	✓
36	H3	Within-domain (translation) motif failure ...	Motif-only AUC (translation)	0.50476	—	—	ratio=0.0048	AUC > 0.5	✓
37	H3	Within-domain (multi_hop_reasoning) motif ...	Motif-only AUC (multi_hop_reasoning)	0.38655	—	—	ratio=-0.1134	AUC > 0.5	×

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#	Hyp	Sub-claim	Primary Metric	Value	95% CI	p-value	Effect Size	Criterion	Met
38	H3	Deviation from domain-mean predicts failures	Domain+graph+deviation AUC (logistic)	0.52949	—	—	ratio=0.0295	AUC > domain+grap...	×
39	H4	FFL hub nodes cause more downstream attr l...	Downstream attr loss median ratio (la...	1.9196	[1.909, 1.934]	$< 10^{-300}$	Cohen's =0.4127	Ratio > 1.5 with ...	✓
40	H4	FFL hub nodes more important than degree-m...	Downstream attr loss median ratio (de...	1.3538	—	—	Cohen's =0.3405	Ratio > 1.0	✓
41	H4	FFL hub nodes more important than random c...	Downstream attr loss median ratio (ra...	9.2267	—	—	Cohen's =0.5425	Ratio > 1.0	✓
42	H4	FFL hub removal causes extreme graph fragm...	Component fragmentation mean ratio (v...	101.6553	—	$< 10^{-300}$	Cohen's =0.1937	Ratio > 10	✓
43	H4	Higher motif participation -> greater abla...	Spearman r (MPI vs downstream attr loss)	0.87680	—	$< 10^{-300}$	ratio=0.8768	r > 0.5 with p < ...	✓
44	H4	Higher motif participation -> greater frag...	Spearman r (MPI vs component fragment...	0.13596	—	7.06×10^{-67}	ratio=0.1360	r > 0 with p < 0.05	✓
45	H4	Effect size of FFL hub ablation (layer-mat...	Cohen's d (downstream attr loss, laye...	0.41275	—	—	Cohen's =0.4127	d > 0.2 (small ef...	✓
46	H4	FFL hub ablation impact in antonym domain	Layer-matched median ratio (antonym)	1.9810	—	$< 10^{-300}$	Cohen's =0.4344	Ratio > 1.0 with ...	✓
47	H4	FFL hub ablation impact in arithmetic domain	Layer-matched median ratio (arithmetic)	2.0351	—	$< 10^{-300}$	Cohen's =0.4979	Ratio > 1.0 with ...	✓
48	H4	FFL hub ablation impact in code_completion...	Layer-matched median ratio (code_comp...	1.8128	—	$< 10^{-300}$	Cohen's =0.3917	Ratio > 1.0 with ...	✓
49	H4	Ablation effect is universal across domains	Number of domains with significant ef...	8.0000	—	—	ratio=1.000	All 8 domains sig...	✓
50	H5	FFL motifs respect layer ordering	Strict layer ordering fraction (3-nod...	1.0000	—	—	ratio=1.000	Fraction > 0.8	✓
51	H5	FFL motifs have coherent sign patterns	FFL coherence fraction	0.58000	—	—	ratio=0.5800	Fraction > 0.5	✓
52	H5	FFL semantic roles vary across domains	Cramer's V (3-node FFL, cross-domain)	0.13000	—	0.001	Cramer's=0.1300	V > 0.3 (moderate)	×
53	H5	Universal 4-node motifs are entirely FFL-d...	4-node FFL containment (all 4 types)	1.0000	—	—	ratio=1.000	100% containment	✓
54	H5	4-node motif 77 cross-domain consistency	Cramer's V (4-node type 77)	0.20000	—	—	Cramer's=0.2000	V > FFL's 0.13	✓
55	H5	4-node motif 80 cross-domain consistency	Cramer's V (4-node type 80)	0.25000	—	—	Cramer's=0.2500	V > FFL's 0.13	✓
56	H5	4-node motif 82 cross-domain consistency	Cramer's V (4-node type 82)	0.30000	—	—	Cramer's=0.3000	V > FFL's 0.13	✓
57	H5	4-node motif 83 cross-domain consistency	Cramer's V (4-node type 83)	0.33000	—	—	Cramer's=0.3300	V > FFL's 0.13	✓
58	H5	FFL layer ordering exceeds random baseline	Random baseline strict ordering fraction	0.45000	—	—	ratio=2.222	FFL ordering >> r...	✓

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#	Hyp	Sub-claim	Primary Metric	Value	95% CI	p-value	Effect Size	Criterion	Met
59	H5	FFL intensity discriminates between capabi...	FFL intensity eta-squared (ANOVA, ite...	0.91700	—	—	eta-squa=0.9170	eta-squared > 0.14	✓

S3 Per-Domain Z-Score Breakdowns

Table 5 presents per-domain Z-score statistics for FFL (030T) and one representative non-FFL motif (021U). The 3-node motif spectrum exhibits degeneracy: 021U, 021C, and 021D Z-scores are identical in magnitude (anti-correlated with 030T).

Table 5: Per-domain Z-score statistics for FFL (030T) and 021U motifs across 200 graphs.

	FFL (030T)			021U (anti-FFL)		
Domain	Mean Z ± SD	Median Z	n	Mean Z ± SD	Median Z	n
Antonym	29.8 ± 9.0	27.4	26	-29.8 ± 9.0	-27.4	26
Arithmetic	57.7 ± 9.6	57.8	26	-57.7 ± 9.6	-57.8	26
Code Completion	46.6 ± 8.9	47.2	24	-46.6 ± 8.9	-47.2	24
Country Capital	47.0 ± 6.4	46.9	26	-47.0 ± 6.4	-46.9	26
Multi Hop Reasoning	38.5 ± 7.2	38.4	24	-38.5 ± 7.2	-38.4	24
Rhyme	48.8 ± 7.6	49.0	24	-48.8 ± 7.6	-49.0	24
Sentiment	59.3 ± 7.8	58.3	24	-59.3 ± 7.8	-58.3	24
Translation	49.8 ± 8.6	49.5	26	-49.8 ± 8.6	-49.5	26
Corpus	47.1 ± 12.1	47.8	200	—	—	—
Cohen's d	3.89			—		

Table 6: Comparison with biological network FFL Z-scores.

Network	FFL Z-score	Network Size	LLM/Bio Ratio
E. coli transcription regulation network	12.7	424	3.71×
S. cerevisiae transcription regulation network	8.5	688	5.55×
C. elegans neural connectivity	10.2	282	4.62×
LLM Mean (200 graphs)	47.1	—	—

S4 Per-Domain Ablation Results

Table 7 presents per-domain ablation results comparing FFL hub node removal against four control types.

Table 7: Per-domain downstream attribution loss: median ratio (hub/control) and significance.

Domain	Degree Matched	Attribution Matched	Layer Matched	Random
Antonym	1.45***	1.03***	1.98***	8.82***
Arithmetic	1.42***	1.00***	2.04***	9.23***
Code Completion	1.29***	1.01***	1.81***	8.32***
Country Capital	1.36***	1.00***	1.94***	9.33***
Multi Hop Reasoning	1.31***	1.03***	1.95***	9.27***
Rhyme	1.37***	1.01***	1.88***	9.74***
Sentiment	1.10***	1.00***	1.65***	6.93***
Translation	1.52***	1.02***	1.88***	11.56***

Dose-Response Analysis. Downstream Attr Loss: Spearman $r = 0.88$. Path Disruption: Spearman $r = 0.13$. Reachability Loss: Spearman $r = 0.03$. Component Fragmentation: Spearman $r = 0.14$.

Corpus Summary. Total FFLs enumerated: 11,185,566. Hub nodes: 20056 (7.94%). Participant nodes: 179193 (70.93%).

Table 8: Corpus-level hub vs. control comparisons (downstream attribution loss).

Control Type	Median Ratio	Wilcoxon p	Cohen's d	n pairs
Degree Matched	1.35	$< 10^{-300}$	0.34	60168
Attribution Matched	1.02	$< 10^{-300}$	0.28	60168
Layer Matched	1.92	$< 10^{-300}$	0.41	59386
Random	9.23	$< 10^{-300}$	0.54	60168

S5 Failure Prediction Extended Results

Table 9 presents all classifier configurations tested for predicting model failures from attribution graph features.

Table 9: Classifier battery: 7 feature sets × 2 classifiers with stratified 5-fold CV.

Feature Set	Classifier	n feat.	AUC	Accuracy	F1
graph stats only	logistic L2	8	0.58	0.73	0.84
domain plus graph	logistic L2	16	0.57	0.72	0.84
full model	logistic L2	21	0.55	0.70	0.83
domain only	logistic L2	8	0.53	0.72	0.84
domain only	random forest	8	0.53	0.72	0.84
domain graph plus dev	logistic L2	23	0.53	0.66	0.80
graph stats only	random forest	8	0.52	0.69	0.82
domain plus graph	random forest	16	0.52	0.69	0.81
full model	random forest	21	0.52	0.66	0.80
full plus dev	random forest	28	0.52	0.68	0.81
full plus dev	logistic L2	28	0.51	0.65	0.78
motif only	logistic L2	5	0.50	0.72	0.83
domain graph plus dev	random forest	23	0.49	0.71	0.83
motif only	random forest	5	0.49	0.66	0.80

Table 10: Within-domain leave-one-out analysis: motif-only vs. graph-stats-only AUC.

Domain	Motif AUC	Graph Stats AUC	n total	Winner
Arithmetic	0.24	0.39	26	Graph Stats
Sentiment	0.19	0.27	24	Graph Stats
Translation	0.50	0.33	26	Motif
Code Completion	0.52	0.41	24	Motif
Country Capital	0.01	0.00	26	Motif
Multi Hop Reasoning	0.39	0.01	24	Motif

Key Pairwise Comparisons. full vs domain graph logistic: mean diff = -0.0202, $p = 0.8650$. full vs domain graph rf: mean diff = -0.0013, $p = 0.5330$. motif vs graph stats logistic: mean diff = -0.0865, $p = 0.9820$.

S6 Corpus-Level Statistical Framework

S6.1 Why Per-Graph BH-FDR Fails

With 50 null models per graph, the minimum achievable p-value is $1/51 \approx 0.0323$. The Benjamini–Hochberg FDR threshold for rank-1 of 800 tests is $0.05/800 = 6.25 \times 10^{-5}$. This resolution gap makes BH-FDR survival mathematically impossible for most tests, regardless of the true effect size.

Table 11: Random forest feature importance (graph statistics features).

Feature	Mean Importance	Std
density	0.102	0.036
n nodes	0.085	0.030
max out degree	0.049	0.030
n edges	0.037	0.033
diameter	0.034	0.026
mean in degree	0.002	0.007
mean out degree	0.002	0.007
n layers	0.000	0.000

S6.2 Corpus-Level Tests

The corpus-level alternative treats each graph's Z-score as one observation and applies standard tests across $n = 200$ graphs:

- One-sample t-test: $t = 55.0$, $p = 2.38 \times 10^{-122}$
- Wilcoxon signed-rank: $W = 0$, $p = 1.44 \times 10^{-34}$
- Sign test: 200/200 positive, $p = 1.24 \times 10^{-60}$
- Cohen's $d = 3.89$ (large effect)

S6.3 Mixed-Effects Model

We fit a random-intercept model: $Z_{ij} \sim \beta_0 + u_j + \varepsilon_{ij}$, where j indexes domains.

- $\beta_0 = 47.2$, 95% CI: [40.5, 53.8]
- $p = 4.93 \times 10^{-44}$
- $\mathrm{Var}(u_j) = 89.23$, $\mathrm{Var}(\varepsilon) = 67.36$
- ICC $= 0.570$

S6.4 Deep Null FDR

With 200 nulls per graph on 15 stratified graphs, 15/60 tests survive BH-FDR at $\alpha = 0.05$.

S6.5 Convergence

Median convergence at $N = 30$ null models. 10% converged by $N = 5$, 70% by $N = 30$, 100% by $N = 50$.

S7 Reviewer Objection-Response Matrix

Table 12 presents anticipated reviewer objections with evidence-based responses and risk assessment.

Table 12: Reviewer Objection-Response Matrix.

#	Objection	Sev.	Evidence-Based Response	Artifacts	Risk
1	*Motifs are just proxies for graph statistics — the high shared variance (93%)...*	4	While 93% of variance is shared, unique motif R-squared=0.0184 with CI [0.0213, 0.0448] strictly excludes zero. Residualized NMI=0.264 (p=0.001) co...	exp_id1_it5	medium
2	*BH-FDR eliminates all per-graph significance — the overrepresentation finding...*	4	FDR eliminates per-graph significance because with only 50 nulls, per-graph p-values have limited resolution. However, the corpus-level signal is o...	exp_id3_it5	low
3	*Only tested on one model (gemma-2-2b) — results may not generalize.*	3	We acknowledge single-model limitation. However, 200 graphs across 8 diverse capability domains (arithmetic, translation, code, reasoning, etc.) pr...	exp_id3_it5, exp_id2_it4	high
4	*Failure prediction completely failed — this undermines the practical value of...*	3	The negative result (best AUC=0.583, motif AUC=0.496) is reported transparently. This strengthens the paper: (a) demonstrates scientific rigor; (b)...	exp_id3_it4	low
5	*3-node motif spectrum degeneracy — with only 1 effective dimension, 'motif an...*	3	Correct — the 3-node spectrum effectively has 1 dimension (FFL vs non-FFL). We address this by: (a) using weighted FFL features (intensity, path do...	exp_id1_it2, exp_id2_it5	medium
6	*Graph-theoretic ablation != real model-level causal intervention — you can't ...*	4	We use careful language: 'graph-theoretic structural importance' not 'causal functional importance'. The ablation quantifies how FFL hubs maintain ...	exp_id2_it4	medium
7	*Small unique R-squared (0.018) — statistically significant doesn't mean pract...*	3	We agree the unique R-squared=0.0184 is small. However: (a) CI excludes 0; (b) residualized NMI=0.264 shows sufficient signal for above-chance doma...	exp_id1_it5	medium
8	*Weak cross-domain semantic consistency (V=0.13) — FFL functional roles are no...*	3	For 3-node FFLs, V=0.13 is indeed weak. However: (a) 4-node types show stronger consistency (V=0.20-0.33); (b) structural properties (100% layer or...	exp_id3_it3, exp_id2_it5, exp_id1_it4	low
9	*Layer-ordering is trivial in layered DAGs — it's an artifact of the graph str...*	2	We agree that 100% strict layer ordering is expected in DAGs with layer assignments. The finding confirms that FFLs respect layer structure but doe...	exp_id3_it3, exp_id2_it5	low
10	*Sample size per domain is small (~25) — insufficient for reliable domain-spec...*	3	Domain sizes range from 24-26 graphs. While small for per-domain inferences, we primarily report corpus-level analyses pooling all 200 graphs. Doma...	exp_id3_it5, exp_id3_it4	medium

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#	Objection	Sev.	Evidence-Based Response	Artifacts	Risk
11	*Attribution graphs may not faithfully represent actual model computation — ga...*	4	Attribution graph faithfulness is an open question. We use Neuronpedia attribution graphs which are state-of-the-art. Our contribution is condition...	exp_id3_it5, exp_id2_it4	high
12	*Confound: graph size drives everything — larger graphs have more motifs, more...*	4	We address this through multiple deconfounding: (a) Z-scores normalize against size-matched null models; (b) unique information decomposition separ...	exp_id3_it5, exp_id1_it5, exp_id2_it4	medium

S8 Limitations and Future Directions

S8.1 Critical Limitations

1. **Single model (gemma-2-2b) — all findings may be model-specific**
   [critical] Mitigation: acknowledged only.
   Future work: Extend to multiple architectures (GPT, Llama, Mistral) and model sizes (7B, 13B, 70B)

2. **Attribution graph faithfulness — unknown whether graphs capture real model computation**
   [critical] Mitigation: acknowledged only.
   Future work: Compare attribution methods; validate with ground-truth circuits from toy models

S8.2 Major Limitations

1. **No model-level causal intervention — graph-theoretic ablation does not prove functional importance**
   [major] Mitigation: partially addressed.
   Future work: Implement actual model-level ablation of FFL hub nodes and measure task accuracy change

2. **3-node motif spectrum degeneracy — effectively 1 dimension, limiting diversity claims**
   [major] Mitigation: partially addressed.
   Future work: Extend to larger motifs (5-node+); develop weighted motif spectrum theory

3. **Small unique motif R-squared (0.018) — most variance shared with graph statistics**
   [major] Mitigation: fully addressed.
   Future work: Investigate non-linear unique information measures; develop less-correlated motif features

4. **BH-FDR eliminates per-graph significance — overrepresentation is corpus-level only**
   [major] Mitigation: fully addressed.
   Future work: Increase null models per graph (100+) to improve per-graph p-value resolution

5. **Failure prediction completely negative — motifs cannot predict model errors**
   [major] Mitigation: fully addressed.
   Future work: Investigate whether finer-grained failure modes are predictable

S8.3 Minor Limitations

1. **Small per-domain sample sizes (~25 graphs per domain)**
   [minor] Mitigation: partially addressed.
   Future work: Collect more attribution graphs per domain; include more capability domains

2. **Weak semantic consistency for 3-node FFLs (V=0.13)**
   [minor] Mitigation: partially addressed.
   Future work: Develop richer semantic characterization methods

3. **Pruning sensitivity — Z-scores depend on pruning threshold (rho=0.29-0.50)**
   [minor] Mitigation: fully addressed.
   Future work: Develop pruning-invariant motif measures; use multiple thresholds in reporting